Störmer's Rule

Mathematical Derivation from First Principles

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1 Problem Statement

We consider a second-order ordinary differential equation of the form

$$y'' = f(x, y), \qquad y(x_0) = y_0, \quad y'(x_0) = v_0, \tag{1}$$

where $y : \mathbb{R} \to \mathbb{R}^n$, $f : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$, and crucially f *does not depend on* y'.

Remark 1 (Physical motivation)**.** *Equation* (1) *describes any purely position-dependent force: gravitational n-body, Keplerian orbits, conservative mechanical systems, and the Störmer problem (charged particle in a magnetic field) that originally motivated the method. For the two-body orbital problem: $f(\mathbf{r}) = -\mu\,\mathbf{r}/|\mathbf{r}|^3$.*

Goal. Advance the solution from x_0 to $x_0 + H$ by subdividing H into M equal sub-steps of size $h = H/M$, in a way that:

1. exploits the absence of y' in f,
2. produces an *even* error series in h (enabling Bulirsch-Stoer Richardson extrapolation), and
3. requires only $M/2$ new evaluations of f per column (half the cost of applying the standard modified midpoint rule to the equivalent first-order system).

2 The Störmer Three-Point Recurrence

2.1 Derivation from Taylor's Theorem

Write the Taylor expansions of y centered at $x_m = x_0 + mh$:

$$y(x_m + h) = y_m + hy'_m + \tfrac{h^2}{2}y''_m + \tfrac{h^3}{6}y'''_m + \tfrac{h^4}{24}y^{(4)}_m + \mathcal{O}(h^5), \tag{2}$$

$$y(x_m - h) = y_m - hy'_m + \tfrac{h^2}{2}y''_m - \tfrac{h^3}{6}y'''_m + \tfrac{h^4}{24}y^{(4)}_m + \mathcal{O}(h^5). \tag{3}$$

Adding (2) and (3) cancels all odd powers:

$$y(x_m + h) + y(x_m - h) = 2y_m + h^2 y''_m + \tfrac{h^4}{12}y^{(4)}_m + \mathcal{O}(h^6). \tag{4}$$

Substituting $y''_m = f(x_m, y_m)$ and rearranging:

$$\boxed{y_{m+1} = 2y_m - y_{m-1} + h^2 f(x_m,\, y_m) + \mathcal{O}(h^4).} \tag{5}$$

This is the **Störmer three-point formula**: given two successive solution values y_{m-1} and y_m, the next value y_{m+1} requires a single evaluation of f and is accurate to $\mathcal{O}(h^4)$ locally.

Remark 2 (No velocity in the recurrence)**.** *Equation* (5) *uses y_{m-1}, y_m, and $f(x_m, y_m)$ only. The velocity y' has been eliminated algebraically. The price is the need for two starting values.*

2.2 The δ-Increment Formulation

Define the position increment $\delta_m := \theta_{m+1} - \theta_m$, the change in the numerical solution over one sub-step. From (5):

$$\begin{aligned}
\delta_m &= y_{m+1} - y_m \\
&= (2y_m - y_{m-1} + h^2 f_m) - y_m \\
&= (y_m - y_{m-1}) + h^2 f_m \\
&= \delta_{m-1} + h^2 f_m.
\end{aligned} \tag{6}$$

This telescoping relation propagates δ forward with only $h^2 f_m$ added at each step. The velocity information is encoded in the initial δ_0.

Starting value from Taylor: The value $y_1 = y(x_0 + h)$ from (2) with $m = 0$:

$$y_1 = y_0 + hv_0 + \tfrac{h^2}{2} f_0 + \mathcal{O}(h^3), \tag{7}$$

so

$$\boxed{\delta_0 = h\, v_0 + \tfrac{h^2}{2}\, f_0.} \tag{8}$$

3 Order Analysis: What "Order p" Means

The three-point recurrence (5) was derived from Taylor's theorem. We now ask precisely: *how fast does the global error shrink as $h \to 0$?* The answer, "order p", refers entirely to powers of h — it is **not** the degree of a polynomial fitted to f.

3.1 The HNW Framework

Following Hairer, Nørsett & Wanner [**?**] (henceforth **HNW**), write the general second-order multistep method for $y'' = f(x, y)$ as

$$\sum_{i=0}^{k} \alpha_i\, y_{n+i} \;=\; h^2 \sum_{i=0}^{k} \beta_i\, f(x_{n+i},\, y_{n+i}), \qquad \text{HNW (III.10, Eq. 10.19)} \tag{9}$$

with scalar sequences $\{\alpha_i\}$ and $\{\beta_i\}$, $\alpha_k \neq 0$. Define the two **characteristic polynomials**

$$\varrho(\zeta) = \sum_{i=0}^{k} \alpha_i \, \zeta^i, \qquad \sigma(\zeta) = \sum_{i=0}^{k} \beta_i \, \zeta^i. \tag{10}$$

These encode the entire method: ϱ governs stability and σ governs accuracy.

The residual when the *exact* solution is substituted into the method is measured by the **linear difference operator** (HNW, Eq. 10.23, p. 467):

$$\mathcal{L}[y,\, x,\, h] \;=\; \varrho(E)\, y(x) \;-\; h^2\, \sigma(E)\, y''(x) \;=\; \sum_{i=0}^{k} \Big[\alpha_i \, y(x+ih) - h^2\, \beta_i \, y''(x+ih) \Big], \tag{11}$$

where E is the shift operator $E\, y(x) = y(x+h)$. Along the exact solution $y''(x+ih) = f\big(x+ih,\, y(x+ih)\big)$, so $\mathcal{L}$ is a pure function of the smooth exact solution y and the step h.

Theorem 1 (HNW Definition 10.2 and Theorem 10.3, p. 467–468)**.** *Method* (9) *is* ***consistent of order*** p *if*

$$\mathcal{L}[y,\, x,\, h] = \mathcal{O}(h^{p+2}) \quad \textit{for every sufficiently smooth } y. \tag{12}$$

Three equivalent conditions (HNW Theorem 10.3, p. 468) are:

(i) Moment form. $\sum_i \alpha_i = 0$, $\sum_i i\, \alpha_i = 0$, *and for* $q = 2, \dots, p{+}1$*:*

$$\sum_{i=0}^{k} \alpha_i \, i^q \;=\; q(q-1) \sum_{i=0}^{k} \beta_i \, i^{q-2}. \tag{13}$$

(ii) Generating-polynomial form.

$$\varrho(e^h) - h^2\, \sigma(e^h) = \mathcal{O}(h^{p+2}) \quad \textit{as } h \to 0. \tag{14}$$

(iii) Root form near $\zeta = 1$.

$$\frac{\varrho(\zeta)}{(\log \zeta)^2} - \sigma(\zeta) = \mathcal{O}\big((\zeta-1)^p\big) \quad \textit{as } \zeta \to 1. \tag{15}$$

Why h^{p+2}, not h^p? The method equates the h^2-scaled forcing $h^2 f$ to the finite-difference $\sum \alpha_i y_{n+i}$, so the entire equation is naturally $\mathcal{O}(h^2)$. The residual $\mathcal{L}$ is therefore already premultiplied by h^2; order p means the *extra* smallness beyond h^2 is h^p more — i.e., h^{p+2} total. The global error, obtained by accumulating $\mathcal{O}(h^{p+2})$ local residuals over $\mathcal{O}(1/h)$ steps and dividing once by the implicit h^2 factor, is $\mathcal{O}(h^p)$ — matching Definition 10.2. This is confirmed by:

Theorem 2 (HNW Theorem 10.6, p. 468 — Global Convergence)**.** *If method* (9) *is stable and of order* p*, and the starting values satisfy* $y(x_j) - y_j = \mathcal{O}(h^{p+1})$ *for* $j = 0, \dots, k-1$*, then*

$$\big|y(x_n) - y_n\big| \le C\, h^p \qquad \textit{for all } 0 \le hn \le \text{const}. \tag{16}$$

3.2 Verification for the $k = 2$ Störmer Method

For the three-point formula (5):

$$\alpha_0 = 1,\ \alpha_1 = -2,\ \alpha_2 = 1; \qquad \beta_0 = 0,\ \beta_1 = 1,\ \beta_2 = 0. \tag{17}$$

Hence $\varrho(\zeta) = \zeta^2 - 2\zeta + 1 = (\zeta - 1)^2$ and $\sigma(\zeta) = 1$ (the center-point quadrature rule for f).

Apply the generating-polynomial test (14). Expanding $\varrho(e^h)$ term by term:

$$\varrho(e^h) = e^{2h} - 2e^h + 1 = h^2 + \frac{h^4}{12} + \frac{h^6}{360} + \cdots \tag{18}$$

(all odd powers cancel because $e^{2h} - 2e^h + 1$ is symmetric under $h \mapsto -h$ after recentering). With $\sigma(\zeta) = 1$:

$$\varrho(e^h) - h^2\sigma(e^h) = \underbrace{h^2}_{\text{cancel}} + \frac{h^4}{12} + \frac{h^6}{360} + \cdots - h^2 = \frac{h^4}{12} + \frac{h^6}{360} + \cdots = \mathcal{O}(h^4). \tag{19}$$

Comparing with condition (12): $h^{p+2} = h^4 \Rightarrow \boxed{p = 2}$. The method is **globally second-order accurate** in h.

The leading residual coefficient $\frac{1}{12}$ is the **error constant** $C = \sigma_2 = \frac{1}{12}$ (HNW Exercise 5, §III.10). It governs the asymptotic error via (HNW Eq. 10.36–10.37, p. 471):

$$\kappa''(x) = \frac{\partial f}{\partial y}\big(x,\, y(x)\big)\, \kappa(x) - \frac{1}{12}\, y^{(4)}(x), \tag{20}$$

where $\kappa(x)$ satisfies $y_n - y(x_n) = \kappa(x_n)\, h^2 + \mathcal{O}(h^3)$. For orbital mechanics, $y^{(4)}$ is the fourth derivative of the position trajectory; a smooth orbit (low eccentricity) keeps $y^{(4)}$ small, making the error constant favorable.

3.3 The Rounding-Error Trap: Why the δ-Form Is Mandatory

The polynomial $\varrho(\zeta) = (\zeta - 1)^2$ has a **double root at** $\zeta = 1$. HNW Definition 10.1 (p. 467) allows roots on the unit circle with multiplicity at most 2 — our method sits exactly on this boundary.

What this means in practice (HNW §III.10, pp. 472–473, Fig. 10.3): a *direct* implementation that stores $y_0, y_1, \ldots$ and applies the recurrence $y_{n+1} = 2y_n - y_{n-1} + h^2 f_n$ accumulates rounding errors that grow without bound as $h \to 0$. The double root causes the companion matrix of the difference equation to be no longer power-bounded.

The **fix** is the δ-increment form (HNW Eq. 10.38′, p. 472):

$$\delta_n - \delta_{n-1} = h^2 f_n, \qquad y_{n+1} - y_n = \delta_n. \tag{21}$$

This splits $\varrho(\zeta)$ as $(\zeta - 1)(\zeta - 1)$ and introduces $\delta_n = y_{n+1} - y_n$ as a new variable. The resulting companion matrix is power-bounded (HNW Eq. 10.30), and rounding errors remain bounded. This is exactly the formulation used in `StormerStep::step()`.

A second pitfall (HNW p. 473, note b): when computing *starting values* $\delta_0, \delta_1, \ldots$, do **not** form the differences $(y_1 - y_0)/h$, $(y_2 - y_1)/h$ from separately computed RK values — that difference is numerically unstable. Instead, accumulate the increments directly from the sub-steps of the RK starting procedure, just as `StormerStep` does internally.

4 Störmer's Modified Midpoint Algorithm

Notation. We introduce θ_m to denote the **numerical approximation** to the true solution $y(x_m)$ at sub-step m, where $x_m = x_0 + mh$. The exact solution is $y(x_m)$; its approximation is θ_m. The algorithm computes a sequence of approximations:

$$\theta_0 \approx y(x_0), \quad \theta_1 \approx y(x_1), \quad \ldots, \quad \theta_M \approx y(x_0 + H).$$

The velocity approximation at the final point is denoted $z_M \approx y'(x_0 + H)$.

Why θ instead of y? In numerical analysis, it is standard to distinguish the *exact* solution $y(x)$ (unknown, continuous) from the *computed* approximation θ_m (known, discrete). The error at step m is $y(x_m) - \theta_m$. This distinction becomes crucial when analyzing convergence and error propagation.

The complete algorithm for M sub-steps of size $h = H/M$ is:

1. **Initialize.**

$$\theta_0 = y_0, \tag{22}$$

$$\delta_0 = h\, v_0 + \tfrac{h^2}{2}\, f(x_0,\, \theta_0). \tag{23}$$

   Set $\theta_1 = \theta_0 + \delta_0$.

2. **Sub-step loop** for $m = 1, \ldots, M-1$:

$$f_m = f(x_0 + mh,\ \theta_m), \tag{24}$$

$$\delta_m = \delta_{m-1} + h^2\, f_m, \tag{25}$$

$$\theta_{m+1} = \theta_m + \delta_m. \tag{26}$$

3. **Final derivative** (velocity estimate at $x_0 + H$):

$$z_M = \frac{\delta_{M-1}}{h} + \frac{h}{2}\, f(x_0 + H,\ \theta_M). \tag{27}$$

The output is $(\theta_M,\ z_M) \approx (y(x_0 + H),\ y'(x_0 + H))$, the numerical approximations to the true position and velocity at $x_0 + H$, with local truncation error $\mathcal{O}(h^2)$.

Total function evaluations of f: one per sub-step $m = 0, \ldots, M$, i.e. $M + 1$ total, but f_0 is supplied from the previous step, so M *new evaluations*.

5 Equivalence to Störmer-Verlet / Leapfrog

Define the half-integer velocity $v_{m+1/2} \coloneqq \delta_m/h$. Rewriting the algorithm in terms of v:

$$v_{1/2} = \frac{\delta_0}{h} = v_0 + \frac{h}{2}\, f_0 \qquad \text{(half-kick)}, \tag{28}$$

$$\theta_{m+1} = \theta_m + h\, v_{m+1/2} \qquad \text{(drift)}, \tag{29}$$

$$v_{m+3/2} = v_{m+1/2} + h\, f_{m+1} \qquad \text{(full kick, } m = 0, \ldots, M-2\text{)}, \tag{30}$$

and the final velocity:

$$z_M = v_{M-1/2} + \frac{h}{2}\, f_M. \tag{31}$$

Equation (31) can be read as: take the half-integer velocity $v_{M-1/2}$, and kick it forward by a half step to return to integer time M. Equivalently:

$$z_M = \frac{v_{M-1/2} + v_{M+1/2}}{2}, \qquad \text{where } v_{M+1/2} = v_{M-1/2} + h\, f_M. \tag{32}$$

So z_M is the arithmetic mean of the bracketing half-integer velocities, a second-order approximation to $y'(x_0 + H)$.

Remark 3 (Symplecticity)**.** *The drift–kick–drift structure* (28)–(30) *is the* Störmer-Verlet *integrator, which is* symplectic *for Hamiltonian systems $H = \frac{1}{2}|v|^2 + V(y)$ with $f = -\nabla V$. It preserves a modified Hamiltonian exactly to all orders. Importantly, it does NOT suffer from secular energy drift.*

6 Even Error Series: Gragg's Theorem

Theorem 3 (Gragg 1965)**.** *Let $y(x)$ be sufficiently smooth. For Störmer's modified midpoint rule applied to $y'' = f(x, y)$ with M sub-steps of $h = H/M$ and the starting/ending conventions* (8)–(27)*, the global errors admit expansions in* even powers *of h only:*

$$y(x_0 + H) - \theta_M = a_2(H)\, h^2 + a_4(H)\, h^4 + a_6(H)\, h^6 + \cdots, \tag{33}$$

$$y'(x_0 + H) - z_M = b_2(H)\, h^2 + b_4(H)\, h^4 + b_6(H)\, h^6 + \cdots, \tag{34}$$

where the coefficients $a_{2k}(H)$, $b_{2k}(H)$ are independent of M.

Sketch of proof (time-reversibility argument). The Störmer recurrence (22)–(27) is *time-reversible*: reversing $h \to -h$ and the sequence of θ_m recovers the original sequence. A time-reversible method cannot accumulate odd powers of h in its global error, because odd-power terms change sign under $h \to -h$ while the exact solution does not. More formally, one expands $\theta_m(h)$ as a power series in h and shows that all odd-power terms cancel identically (see Hairer, Nørsett & Wanner, Vol. I, §II.8 for the complete argument).

Corollary 1 (Richardson extrapolation efficiency)**.** *Because the error series* (33) *contains only h^2, h^4, …, a single Richardson extrapolation step using results at h and $h/2$ eliminates the h^2 term and yields $\mathcal{O}(h^4)$ accuracy. In general, k extrapolation steps yield $\mathcal{O}(h^{2k+2})$ accuracy from $\mathcal{O}(h^2)$ base integrations. This is the foundation of the Bulirsch-Stoer method.*

7 Richardson Extrapolation Setup (Bulirsch-Stoer)

Evaluate the modified midpoint rule with sub-step counts $M_1 < M_2 < M_3 < \cdots$ (the Bulirsch-Stoer sequence: $\{2, 4, 6, 8, 12, 16, \ldots\}$), giving approximations $T_{j,0} \coloneqq \theta_{M_j}(H)$.

Define the extrapolation table recursively:

$$T_{j,k} = T_{j,k-1} + \frac{T_{j,k-1} - T_{j-1,k-1}}{\left(M_j/M_{j-k}\right)^2 - 1}, \qquad k = 1, 2, \ldots, j. \tag{35}$$

Each extrapolation step eliminates the next order in h^2. The diagonal $T_{k,k}$ converges to $y(x_0 + H)$ as k increases.

Remark 4 (Computational cost vs. accuracy)**.** *For the first-order system $[y', f(x, y)]^T$ of dimension $2n$, the modified midpoint rule needs M evaluations of f per column. Störmer's rule needs only $M/2$ evaluations per column (the NR3 code uses $n_{\mathit{eff}} = M/2$ leapfrog steps of $h_{\mathit{eff}} = 2h = 2H/M$, as shown below) for the same total interval H. Since f can be expensive (e.g., gravitational force with many bodies), this factor of two is significant.*

8 Analysis of the NR3 Implementation

8.1 Parameterization

NR3's `StepperStoerm::dy()` uses:

- `nstep` $= M$ (from Bulirsch-Stoer sequence $\{2, 4, 6, 8, \ldots\}$).
- `h` $= H/M$ (unit sub-step).
- `h2` $= 2h = 2H/M$ (actual loop step, double the unit sub-step).
- `nstp2` $= M/2$ (number of loop iterations).

8.2 Algorithm in NR3 variables

Initialization:

$$\begin{aligned} \text{ytemp}[i] &\leftarrow y[i] & \qquad \theta_0 &= y_0, \\ \text{ytemp}[n+i] &\leftarrow y'[i] + h \cdot f_0 & \qquad v^* &= v_0 + h\, f_0 = v_{1/2} \cdot \underbrace{2}_{h_{\text{eff}}/h}. \end{aligned}$$

In other words, the stored "velocity" `ytemp`$[n+i]$ is the half-integer velocity $v_{1/2} = v_0 + (h/2)\, f_0$ *scaled by 2*: `ytemp`$[n+i] = 2\, v_{1/2}$. Then the drift step below with factor $h_{\text{eff}} = 2h$ is:

$$\theta_1 = \theta_0 + h_{\text{eff}} \cdot \text{ytemp}[n+i]/2 = \theta_0 + h_{\text{eff}}\, v_{1/2} = \theta_0 + 2h \left(v_0 + \tfrac{h}{2}\, f_0\right) = \theta_0 + 2h\, v_0 + h^2\, f_0.$$

Alternatively, viewing the NR3 scheme directly as Störmer-Verlet with step $h_{\text{eff}} = 2h$:

$$\text{Loop } m = 1, \ldots, n_{\text{eff}} : \quad \begin{cases} \theta_m = \theta_{m-1} + h_{\text{eff}}\, v_{m-1/2}, \\ v_{m+1/2} = v_{m-1/2} + h_{\text{eff}}\, f(x_0 + m\, h_{\text{eff}},\ \theta_m), \quad m < n_{\text{eff}}. \end{cases} \tag{36}$$

Final velocity: $z = v_{n_{\text{eff}}-1/2} + (h_{\text{eff}}/2)\, f(x_0 + H, \theta_{n_{\text{eff}}})$.

8.3 Cost comparison

Method	State dimension	Evals of f per column	Error order
Midpoint (first-order 2n-system)	$2n$	M	$\mathcal{O}(h^2)$
Störmer (NR3)	n	$M/2$	$\mathcal{O}(h_{\text{eff}}^2)$

Both methods embed in a Bulirsch-Stoer extrapolation with an even-power error series and converge to the same accuracy. Störmer uses half the evaluations of f at the cost of doubling the effective sub-step size, which the extrapolation corrects.

9 Is Inheritance Necessary?

NR3's design uses `struct StepperStoerm : StepperBS<D>` *solely* for code reuse of the Bulirsch-Stoer extrapolation machinery (`odeint` driver, column sequence, Richardson table, stepsize control). The method `dy()` is entirely independent of the base class algorithms; it overrides only the *sub-step integration* while reusing everything else.

From first principles, the minimal reusable components are:

1. `StormerStep`: executes n_{eff} leapfrog sub-steps over a given interval H. No base class required. Template on `DerivativeType` and `ContainerT`.
2. **Error estimation**: step-doubling (compare n_{eff} steps vs. $2\,n_{\text{eff}}$ steps) or Richardson extrapolation as a separate, composable component.
3. **Step-size control**: compose with the existing `ComputePIStepSize` / `PIStepSizeControl` classes already present in the library.

Composition over inheritance, exactly consistent with how `CalculateNewYAndError` and `IntegrateWithPIControl` were designed.

10 Harmonic Oscillator Test Case

The standard test for Störmer's rule is the harmonic oscillator:

$$y'' = -\omega^2 y, \qquad y(0) = 1, \quad y'(0) = 0, \quad \omega = 1. \tag{37}$$

Exact solution: $y(t) = \cos t$, $y'(t) = -\sin t$, with conserved energy:

$$E(t) = \tfrac{1}{2}\big(y'(t)\big)^2 + \tfrac{1}{2}\big(y(t)\big)^2 = \tfrac{1}{2}. \tag{38}$$

Störmer-Verlet (symplectic) conserves a modified energy exactly, so the energy error remains bounded for all time (no secular drift), unlike non-symplectic methods.

Expected convergence. For a single step of size $H = 0.5$ with n_{eff} leapfrog sub-steps of $h_{\text{eff}} = H/n_{\text{eff}}$:

$$|y(H) - \theta_M| \approx \tfrac{H^3}{24}\,|y'''(0)| \cdot h_{\text{eff}}^2 = \tfrac{H^3\omega^3}{24} \cdot h_{\text{eff}}^2 = \mathcal{O}(h_{\text{eff}}^2).$$

Doubling n_{eff} (halving h_{eff}) should reduce the error by a factor of 4. This is the convergence test in the unit tests.

11 Numerov's Method: The Implicit Upgrade

Section 3 showed that the three-point stencil (5) achieves order 2 because $\varrho(e^h) - h^2\sigma(e^h) = \mathcal{O}(h^4)$ with $\sigma(\zeta) = 1$. The same stencil on the left-hand side, but a richer choice of σ, can reach $\mathcal{O}(h^6)$ — i.e. order 4 — with *no extra history.*

11.1 Notation Bridge: HNW Polynomials vs NR3/Our Variables

The analysis in §3 introduced Hairer–Nørsett–Wanner (HNW) notation. Here is a translation table bridging their polynomial formalism with the discrete variables used earlier in this document and in NR3:

Symbol	**Meaning**	**Where it appears**
$\varrho(\zeta)$	First characteristic polynomial $\sum_{i=0}^{k} \alpha_i\zeta^i$	HNW Eq. 10.20, p. 467 (stability)
$\sigma(\zeta)$	Second characteristic polynomial $\sum_{i=0}^{k} \beta_i\zeta^i$	HNW Eq. 10.20, p. 467 (accuracy)
ζ	Formal variable; $E = \zeta$ is the shift operator: $Ey_n = y_{n+1}$	HNW §III.1
α_i	Coefficients of LHS stencil (position weights)	Eq. (9)
β_i	Coefficients of RHS quadrature (force weights)	Eq. (9)
k	Step number (history length k) Method uses $y_n, \ldots, y_{n+k}$	HNW §III.10
h	Step size	Same throughout
H	Total interval (macro-step)	§4
y_n	Exact solution at $x_n = x_0 + nh$	Continuous $y(x_n)$
θ_n	**Computed** approximation to y_n	§4
f_n	Force $f(x_n, y_n)$ or $f(x_n, \theta_n)$	§4
δ_n	Position increment $\theta_{n+1} - \theta_n$	Eq. (6)
$v_{1/2}$	Half-integer velocity (leapfrog)	Eq. (28)

Connection to our earlier notation. In §2–4 we wrote the method as

$$\theta_{m+1} - 2\theta_m + \theta_{m-1} = h^2 f_m.$$

In HNW polynomial form this is

$$\varrho(\zeta) = \zeta^2 - 2\zeta + 1 = (\zeta - 1)^2, \qquad \sigma(\zeta) = 1.$$

The coefficients are $(\alpha_0, \alpha_1, \alpha_2) = (1, -2, 1)$ and $(\beta_0, \beta_1, \beta_2) = (0, 1, 0)$. The shift operator E acts as $E\theta_m = \theta_{m+1}$, so the compact operator form is $\varrho(E)\theta_m = h^2\sigma(E)f_m$.

NR3 notation. NR3 uses 'nstep' for the number of sub-steps M, 'h' for H/M, 'h2' for $2h$, and 'nstp2' for $M/2$. Their 'ytemp[n+i]' stores the half-integer velocity $v_{m+1/2}$ scaled by 2 (see §**??**).

11.2 Derivation via the Order Condition

Keep $\varrho(\zeta) = \zeta^2 - 2\zeta + 1$ and seek $\sigma(\zeta) = \beta_0 + \beta_1\zeta + \beta_2\zeta^2$ (symmetric: $\beta_0 = \beta_2$) such that $\varrho(e^h) - h^2\sigma(e^h) = \mathcal{O}(h^6)$.

From (19), $\varrho(e^h) = h^2 + \frac{h^4}{12} + \frac{h^6}{360} + \cdots$. We need $h^2\sigma(e^h)$ to match through the h^5 term. Expanding with $\beta_0 = \beta_2 \equiv \beta$, $\beta_1 \equiv \gamma$:

$$h^2\sigma(e^h) = h^2\big(\beta e^{2h} + \gamma e^h + \beta\big) = h^2\big[(2\beta + \gamma) + 2h\beta + h^2(2\beta + \tfrac{\gamma}{2}) + \cdots\big].$$

Matching through h^5 against $\varrho(e^h)$ forces $2\beta + \gamma = 1$ (the h^2 term), and solving the system for the h^4 match gives $\beta = \frac{1}{12}$, $\gamma = \frac{10}{12}$. Hence:

$$\sigma(\zeta) = \frac{\zeta^2 + 10\zeta + 1}{12}. \tag{39}$$

The resulting method is (HNW Eq. 10.12, p. 464; B. Numerov 1924):

$$\boxed{y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{12}\Big(f_{n+1} + 10\, f_n + f_{n-1}\Big),} \tag{40}$$

called **Numerov's method**. The only difference from (5) is the f_{n+1} term on the right: one additional force evaluation at the new position.

11.3 Order Verification

Apply the generating-polynomial test (14):

$$\begin{aligned} h^2\sigma(e^h) &= \frac{h^2}{12}\big(e^{2h} + 10e^h + 1\big) \\ &= h^2 + \frac{h^4}{12} + \frac{h^6}{144} + \cdots \end{aligned} \tag{41}$$

and recall $\varrho(e^h) = h^2 + \frac{h^4}{12} + \frac{h^6}{360} + \cdots$. Subtracting:

$$\varrho(e^h) - h^2\sigma(e^h) = \Big(\frac{1}{360} - \frac{1}{144}\Big)h^6 + \mathcal{O}(h^8) = -\frac{1}{240}\,h^6 + \mathcal{O}(h^8) = \mathcal{O}(h^6). \tag{42}$$

Comparing with (12): $h^{p+2} = h^6 \Rightarrow \boxed{p = 4}$. **Numerov's method is globally fourth-order accurate.**

The error constant is $C = \sigma_3^* = 0$ (Table 10.2, HNW p. 464) — the leading error is $-h^6/240 \cdot y^{(6)}$ per step, contributing:

$$\kappa''(x) = \frac{\partial f}{\partial y}\,\kappa - \frac{1}{240}\,y^{(6)}(x) \qquad \big(\text{global error: } y_n - y(x_n) = \kappa(x_n)\,h^4 + \mathcal{O}(h^5)\big). \tag{43}$$

11.4 The Order-Barrier Context

HNW Theorem 10.4 (p. 468) states that for a stable k-step Störmer method:

$$p \le k + 2 \ \text{ if } k \text{ is even}, \qquad p \le k + 1 \ \text{ if } k \text{ is odd}. \tag{44}$$

For $k = 2$ (our stencil): $p_{\max} = 4$. Numerov saturates this bound — it is the *highest-order stable method* on the three-point stencil. The explicit method (order 2) leaves two orders on the table.

11.5 PECECE Implementation

Numerov's method is implicit: y_{n+1} appears on both sides via $f_{n+1} = f(x_{n+1}, y_{n+1})$. For nonlinear f (e.g. gravitational force), we solve using **PECECE** (Predict–Evaluate–Correct–Evaluate–Correct–Evaluate):

1. **Predict.** Use the explicit Störmer formula as first guess:
$$y_{n+1}^{(P)} = 2y_n - y_{n-1} + h^2 f_n. \tag{45}$$
2. **Evaluate.** $f_{n+1}^{(P)} = f\left(x_{n+1},\, y_{n+1}^{(P)}\right)$.
3. **Correct.**
$$y_{n+1}^{(1)} = 2y_n - y_{n-1} + \frac{h^2}{12}\left(f_{n+1}^{(P)} + 10f_n + f_{n-1}\right). \tag{46}$$
4. **Evaluate.** $f_{n+1}^{(1)} = f\left(x_{n+1},\, y_{n+1}^{(1)}\right)$.
5. **Correct again.**
$$y_{n+1} = 2y_n - y_{n-1} + \frac{h^2}{12}\left(f_{n+1}^{(1)} + 10f_n + f_{n-1}\right). \tag{47}$$
6. **Evaluate.** $f_{n+1} = f(x_{n+1},\, y_{n+1})$ (stored for next step).

Velocity. Numerov delivers positions. Velocity at integer time is recovered via the leapfrog formula (order 2 in h):
$$v_{n+1} = \frac{y_{n+1} - y_n}{h} + \frac{h}{2}\, f_{n+1}. \tag{48}$$
Position accuracy is $\mathcal{O}(h^4)$; velocity accuracy is $\mathcal{O}(h^2)$. For applications where velocity precision matters as much as position (e.g. orbit determination), the velocity can be improved via central differences: $v_n \approx (y_{n+1} - y_{n-1})/(2h) + \mathcal{O}(h^2)$ — same order.

Order of PECECE. With an order-2 predictor and order-4 corrector, PECE alone is order 3: the predictor error $\mathcal{O}(h^2)$ contaminates $f_{n+1}^{(P)}$ by $\mathcal{O}(h^2)$, adding $\frac{h^2}{12}\cdot\mathcal{O}(h^2) = \mathcal{O}(h^4)$ per step to the corrector residual, giving $\mathcal{O}(h^3)$ global error. The second correction (PECECE) reduces the contamination to $\mathcal{O}(h^6)$ per step — below the corrector's own leading error of $\mathcal{O}(h^6)$ — so the full order-4 accuracy is recovered.

11.6 Cost Comparison

Method	Order p	f evals/step	Error constant C
Explicit Störmer ($k = 2$)	2	1	$\sigma_2 = \frac{1}{12}$
Numerov PECECE ($k = 2$)	4	3	$-\frac{1}{240}$

Halving h reduces error by $2^2 = 4$ for explicit Störmer vs. $2^4 = 16$ for Numerov — a $4\times$ gain per step halving. The cost of 3 f-evals vs. 1 is recovered after the first tolerance tightening: to reach the same accuracy, Numerov needs $\approx 2.5\times$ fewer steps than explicit Störmer (since $4^{1/4}/2^{1/2} \approx 1.41$ ratio in step size for equal error).

11.7 Starting Procedure

Numerov is a two-step method: it needs y_0 and y_1 to start. Given initial conditions (y_0, v_0), one accurate starting step is required. Theorem 2 demands $y(x_1) - y_1 = \mathcal{O}(h^5)$ for the method to retain its full order-4 convergence. Options:

- **Taylor expansion:** $y_1 = y_0 + hv_0 + \frac{h^2}{2} f_0 + \frac{h^3}{6} f_0' + \frac{h^4}{24} f_0''$ (requires derivative information, gives $\mathcal{O}(h^5)$, satisfies requirement).
- **Runge-Kutta-Nyström order 4:** one step of a 4th-order Nyström method (HNW Table 14.2, p. 285) gives $\mathcal{O}(h^5)$ accuracy, at the cost of 3–4 f-evaluations for the single startup step.
- **Two steps of explicit Störmer** (order 2): gives only $\mathcal{O}(h^3)$, limiting global convergence to $\mathcal{O}(h^2)$. Sufficient only for testing.

`NumerovStep::startup()` uses one step of the leapfrog (Störmer-Verlet) with n_{eff} substeps as a high-accuracy starter: for large n_{eff}, this approximates the exact solution to whatever precision is needed. In production, supply a 4th-order starter.

12 Astrodynamics Context: Cowell's Method, Encke's Method, and Perturbations

In orbital mechanics, the equations of motion for a satellite under perturbations are:

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r} + \mathbf{a}_p \tag{49}$$

where $\mathbf{a}_p$ is the sum of all perturbing accelerations. Two main approaches exist for numerical integration:

12.1 Cowell's Method (Direct Integration)

Cowell's method [?, ?, ?] directly integrates the total acceleration:

$$\dot{\mathbf{r}} = \mathbf{v} \tag{50}$$

$$\dot{\mathbf{v}} = -\frac{\mu}{r^3}\mathbf{r} + \mathbf{a}_p \tag{51}$$

Advantages:

- Simplicity of formulation and implementation
- Any number of perturbations can be handled simultaneously
- Obtains $\mathbf{r}$ and $\mathbf{v}$ directly

Disadvantages:

- Near large attracting bodies, smaller steps are required (affects time and roundoff error)
- Approximately 10× slower than Encke's method [?]
- Roundoff error accumulates rapidly in double-precision for interplanetary flights with small step sizes

12.2 Encke's Method (Osculating Orbit)

Encke's method [**?**, **?**] integrates the *deviation* from a reference (osculating) two-body orbit rather than the total state. Let $\rho(t)$ be the osculating orbit position (the path the satellite would follow if perturbations vanished at epoch t_0). The deviation $\delta\mathbf{r} = \mathbf{r} - \rho$ satisfies:

$$\delta\ddot{\mathbf{r}} = \mathbf{a}_p + \frac{\mu}{\rho^3}\left[(1-2q)^{-3/2}\mathbf{r} - \delta\mathbf{r}\right] \tag{52}$$

where $q = \delta\mathbf{r}\cdot(2\mathbf{r} - \delta\mathbf{r})/r^2$.

Advantages:

- Faster than Cowell's method (deviations change slowly)
- Better accuracy for small perturbations
- Less accumulation of roundoff error

Disadvantages:

- More complex formulation
- Requires *rectification* (resetting the osculating orbit) when $\delta r/r$ exceeds a tolerance
- The term $(1-2q)^{-3/2}$ requires careful numerical handling (difference of nearly equal quantities)

12.3 Störmer/Numerov vs. Cowell for Orbital Mechanics

Our `StörmerStep` and `NumerovStep` implementations are **integration methods**, while Cowell's and Encke's are **formulation frameworks**. The relationship:

Formulation	Integration Method	Use Case
Cowell's	Störmer (order 2) or Numerov (order 4)	Direct integration of $y'' = f(x,y)$
Cowell's	RK4, Adams-Moulton	General first-order reduction
Encke's	Störmer/Numerov	Deviation from osculating orbit

Why Numerov outperforms standard Cowell integration:

1. **Order 4 accuracy** vs. order 2 for explicit Störmer or standard RK4 on first-order form
2. **Same stencil** (3 points) as explicit Störmer, but implicit correction achieves higher order
3. **Symplectic-like** energy behavior for near-Keplerian orbits (nearly circular)
4. **Fewer function evaluations** than equivalent RK4: 3 f-evals/step vs. 4 for RK4

Bate et al. [**?**] note that for orbital problems, the **Gauss-Jackson** (sum-squared, Σ^2) method is "one of the best, and most used, numerical integration methods for trajectory problems of the Cowell and Encke type." The Gauss-Jackson method is also a multistep predictor-corrector for second-order equations, similar in spirit to Numerov but with higher order and specialized for orbital mechanics.

12.4 Common Perturbations in Astrodynamics

Based on Curtis [?], Hintz [?], and Bate [?], the primary perturbations affecting Earth satellites are:

1. **Earth Oblateness (J_2)**: The Earth's equatorial bulge causes secular variations in Ω (RAAN) and ω (argument of perigee). The nodal regression rate is:

$$\dot{\Omega} = -\frac{3J_2}{2}\sqrt{\frac{\mu}{a^3}}\left(\frac{R_\oplus}{a(1-e^2)}\right)^2 \cos i \tag{53}$$

2. **Atmospheric Drag**: Significant for LEO satellites. The acceleration is:

$$\mathbf{a}_d = -\frac{1}{2}\rho v_{rel}^2 \frac{C_D A}{m} \frac{\mathbf{v}_{rel}}{|\mathbf{v}_{rel}|} \tag{54}$$

   where ρ is atmospheric density, $C_D \approx 2.0$–2.2 for typical satellites.

3. **Third-Body Gravity** (Moon, Sun): The lunar perturbation acceleration on an Earth satellite is:

$$\mathbf{a}_p = \mu_M \left(\frac{\mathbf{r}_{ms}}{r_{ms}^3} - \frac{\mathbf{r}_{Me}}{r_{Me}^3}\right) \tag{55}$$

   where $\mathbf{r}_{ms}$ is the vector from Moon to satellite, $\mathbf{r}_{Me}$ from Moon to Earth.

4. **Solar Radiation Pressure**: For spacecraft with large area-to-mass ratio:

$$\mathbf{a}_{SRP} = -p_{SR}\frac{A}{m}\cos\theta\,\hat{\mathbf{s}} \tag{56}$$

   where $p_{SR} \approx 4.5 \times 10^{-6}$ N/m^2 at 1 AU.

Special vs. General Perturbations:

- **Special perturbations**: Direct numerical integration (Cowell, Encke) — computes specific trajectories for given initial conditions.
- **General perturbations**: Analytic integration of series expansions (variation of parameters, Lagrange planetary equations) — provides analytical insight but is more complex.

13 Summary

1. Störmer's rule exploits $y'' = f(x, y)$ (no y' in f) via the three-point recurrence $y_{m+1} = 2y_m - y_{m-1} + h^2 f_m$.
2. In δ-form: $\delta_m = \delta_{m-1} + h^2 f_m$, with starting value $\delta_0 = hv_0 + \frac{h^2}{2} f_0$. This form is mandatory to avoid numerical instability from the double root $\varrho(\zeta) = (\zeta - 1)^2$ (Section 3.3).
3. The algorithm is equivalent to Störmer-Verlet (leapfrog), which is symplectic for Hamiltonian systems.

4. "Order p" means global error $\mathcal{O}(h^p)$ in step size h (not polynomial degree). The explicit method is order 2; Numerov's implicit method (40) saturates the order barrier for the two-step stencil at order 4.

5. Gragg's theorem guarantees an even-power error series in h, enabling efficient Bulirsch-Stoer Richardson extrapolation.

6. NR3's `StepperStoerm` implements the $k = 2$ explicit Störmer in one-step extrapolation mode (HNW §II.14, Eq. 14.32) — the simplest special case of the general multistep family (HNW §III.10).

7. Numerov PECECE (Section 11) achieves order 4 on the same three-point stencil at 3 f-evaluations/step vs. 1 for explicit Störmer. Implemented as `NumerovStep`.

References.

- E. Hairer, S. P. Nørsett, G. Wanner, *Solving Ordinary Differential Equations I: Nonstiff Problems*, 2nd ed. (Springer, 1993). Key sections: §II.14 (Nyström methods, extrapolation for $y'' = f(x, y)$), §III.10 (multistep Störmer methods, pp. 461–473).

- W. B. Gragg, "On extrapolation algorithms for ordinary initial value problems," *SIAM J. Numer. Anal.* **2** (1965) 384–403.

- B. Numerov, "A method of extrapolation of perturbations," *Mon. Not. R. Astron. Soc.* **84** (1924) 592–601.

- W. H. Press et al., *Numerical Recipes* 3rd ed. (Cambridge, 2007), §17.4 (`StepperStoerm`: the $k = 2$ one-step extrapolation case).

- C. Störmer, *Sur les trajectoires des corpuscules électrisés*, *Arch. Sci. Phys. Nat.* **24** (1907) 5–18.

R. R. Bate, D. D. Mueller, J. E. White, *Fundamentals of Astrodynamics* (Dover, 1971). §9.2 (Cowell's Method), §9.3 (Encke's Method), §9.6 (Numerical Integration Methods including Gauss-Jackson).

H. D. Curtis, *Orbital Mechanics for Engineering Students*, 2nd ed. (Elsevier, 2005). Ch. 10 (Orbital Perturbations): §10.2 (Cowell), §10.3 (Encke), §10.4 (Atmospheric Drag), §10.5 (Gravitational Perturbations).

G. R. Hintz, *Orbital Mechanics and Astrodynamics* (Springer, 2015). Ch. 5: §5.2 (Cowell's Method), §5.3 (Encke's Method), §5.5 (Integration Schemes), §5.7 (Analytic Formulation of Perturbations).